



## AI Playbook and Toolkit for Public Health Departments

# Workforce Strategy and AI Role Design

EXE 440

AI changes the skills, roles, and collaboration patterns needed in public health agencies. It does not eliminate the need for public health expertise. Instead, it increases the need for staff who can translate across program, data, technology, privacy, legal, equity, procurement, and community perspectives. Leaders need a workforce strategy that prepares staff to use, review, govern, and improve AI-supported workflows. This module helps leaders identify AI-related roles and competencies. It emphasizes the need for role-based training, AI bridge professionals, governance participants, technical reviewers, program owners, data stewards, and frontline users. The module also addresses workload, job redesign, deskilling risk, upskilling, and equitable access to training.

|                   |                                          |
|-------------------|------------------------------------------|
| Level             | advanced/leadership                      |
| Primary track     | Public Health Executive Leadership Track |
| Appears in tracks | governance-security                      |

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## Learning Objectives

- Identify workforce roles needed for responsible AI adoption in public health.
- Distinguish between AI users, reviewers, implementers, governance participants, and accountability owners.
- Develop role-based competency expectations for AI-supported work.
- Recognize workforce risks such as deskilling, uneven access to training, and unclear role boundaries.
- Produce a workforce readiness and role design plan.

## Module Overview

AI changes the skills, roles, and collaboration patterns needed in public health agencies. It does not eliminate the need for public health expertise. Instead, it increases the need for staff who can translate across program, data, technology, privacy, legal, equity, procurement, and community perspectives. Leaders need a workforce strategy that prepares staff to use, review, govern, and improve AI-supported workflows.

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## Why This Topic Matters for Public Health AI

Public health AI depends on people who understand both technology and context. A model may generate a summary, but an epidemiologist must judge whether the summary is meaningful. A vendor may describe a product, but an informatician must assess interoperability and data flow. A leader may approve a use case, but governance participants must ensure that privacy, equity, security, and legal concerns have been addressed.

Workforce strategy helps agencies move beyond ad hoc training. It defines which staff need baseline literacy, which staff need deeper technical skills, which staff need governance training, and which roles require ongoing competency development. It also helps agencies avoid concentrating AI knowledge in a small group while leaving most staff dependent on informal guidance.

## Public Health Example

A public health agency implements several AI tools: a document assistant, a case summarization tool, a data quality monitor, and a predictive model for resource planning. Staff quickly discover that no single role owns all the knowledge needed. Program staff understand workflow, IT understands integration, privacy understands data restrictions, and analysts understand validation. However, there is no role that connects these perspectives consistently.

The agency creates an AI bridge role to support use case development, governance preparation, workflow mapping, vendor questioning, training, and implementation review. The role does not replace technical specialists or program leaders. It helps them work together more effectively.

## Technical and Governance Deep Dive

An AI workforce strategy should define role groups. General staff may need baseline AI literacy, approved-use guidance, and prompt safety. Frontline users need workflow-specific training and escalation expectations. Supervisors need to support review, feedback, and documentation. Program owners need to understand accountability and performance monitoring. Technical staff need deeper knowledge of data, architecture, validation, and security. Governance members need enough shared language to review proposals consistently.

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Competency expectations should be practical. Staff do not all need to become data scientists. However, many need to recognize when AI output is unsupported, when sensitive data should not be entered, when a use case exceeds approved boundaries, and when to escalate. Technical and informatics staff may need competencies in data quality, interoperability, model evaluation, RAG, security, and lifecycle monitoring.

Role design should also address workload and equity. AI review and governance work require time. If these responsibilities are added informally to already overloaded staff, governance may become superficial. Leaders should identify protected time, training resources, role descriptions, and career pathways for staff who support responsible AI.

## Risks, Failure Modes, and Guardrails

One risk is assuming that AI reduces workforce needs. In practice, responsible AI often requires new review, training, documentation, and monitoring work. A guardrail is to include workforce impact in every business case and implementation plan.

Another risk is uneven access to AI skills. Staff in better-resourced programs may receive training first, while others fall behind. A guardrail is to create a role-based training plan that includes all relevant staff groups and supports equitable participation.

## Application to AI-Supported Workflows

Generative AI requires staff who can review outputs and manage source accuracy. Predictive analytics requires staff who can interpret performance and understand limitations. Agentic workflows require staff who can oversee permissions, actions, escalation, and rollback. Workforce strategy should connect each AI type to the roles needed to use it safely.

## Reflection Questions

Which current or proposed AI use case in your agency raises this leadership or governance issue most clearly?

Who currently has authority to make decisions about this issue, and is that authority documented?

What evidence would governance or leadership need before approving the use case?

Which staff, partners, or communities would be affected if this issue were handled poorly?

What policy, process, or artifact should be created or updated after this module?

## Practical Exercise

Create a role-based AI workforce readiness plan. Include role groups, competencies, required training, responsibilities, decision authority, and support needs.

## Before You Begin

Choose a real or realistic public health AI use case. Document assumptions, unresolved questions, and which agency roles would need to review the artifact before it is adopted. The exercise should produce a work product that could support governance review, leadership discussion, implementation planning, or policy development.

## Expected Artifact or Evidence

Workforce readiness plan; role-based competency matrix; training assignment matrix; AI bridge role description.

## Expected Assignment

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- Workforce readiness plan; role-based competency matrix; training assignment matrix; AI bridge role description.

## Knowledge Check

- 1. What is the strongest reason this module topic belongs in AI leadership and governance training?
- 2. Which practice best supports accountable public health AI governance?
- 3. When should an AI use case be escalated for leadership or governance review?
- 4. What is weak evidence of completion for a leadership and governance module?
- 5. What should leaders do when an AI use case has meaningful benefits but unresolved risks?

## References and Resources

- NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0):  
<https://www.nist.gov/itl/ai-risk-management-framework>
- NIST Generative AI Profile: <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
- NIST Privacy Framework
- CDC Public Health Data Strategy and Data Modernization resources:  
<https://www.cdc.gov/public-health-data-strategy/php/about/index.html>
- WHO Ethics and Governance of Artificial Intelligence for Health:  
<https://www.who.int/publications/i/item/9789240029200>
- OMB Memorandum M-24-10 on federal AI governance, risk management, and innovation, where relevant to public-sector governance models
- Agency strategic plans, procurement policies, privacy policies, cybersecurity policies, records policies, and equity plans